CLINICAL · NOSE & SINUSES

Acute rhinosinusitis and orbital complications

Most acute rhinosinusitis is a cold that has reached the sinuses, and most of it settles without antibiotics. The rhinologist's real job is to recognise the minority that turn bacterial, and the few that break out of the sinuses into the orbit, the bone or the brain. A swollen eye in a child with a cold is an emergency until the orbit has been assessed.

AT A GLANCE

- Viral first; bacterial if over 10 days or worsening.
- Any eye swelling is urgent: check vision.
- Use Chandler stages, CT and IV treatment.
- Drain abscesses.
- Post-viral ARS: worse after 5 days or beyond 10; bacterial needs 3 of 5 EPOS features; antibiotics are overused.

1 CHAPTER ONE Overview

A cold that reaches the sinuses.

1.1 Definition

Inflammation of the nose and paranasal sinuses. The term *rhinosinusitis* is preferred, because the nasal mucosa is always involved too.

1.2 Classification by time

CLASSIFICATION OF RHINOSINUSITIS

TYPE	DURATION
Acute	Under 4 weeks, with complete resolution
Subacute	4 to 12 weeks
Chronic	12 weeks or more

TYPE	DURATION
Recurrent acute	4 or more episodes a year, each lasting 7 to 10 days or more, with full recovery between
Acute on chronic	Sudden worsening of chronic disease, returning to baseline

1.3 Symptoms

Nasal blockage, anterior or posterior discharge, facial pain or pressure, and reduced smell, with cough, fever, fatigue, dental pain, halitosis, headache or ear fullness.

2 CHAPTER TWO Applied anatomy

The sinuses and their drainage.

2.1 Drainage

The frontal, anterior ethmoid and maxillary sinuses drain into the middle meatus through the *ostiomeatal complex*; the posterior ethmoids drain into the superior meatus and the sphenoid into the sphenoethmoidal recess. Mucociliary clearance in each sinus follows a fixed path towards its natural ostium.

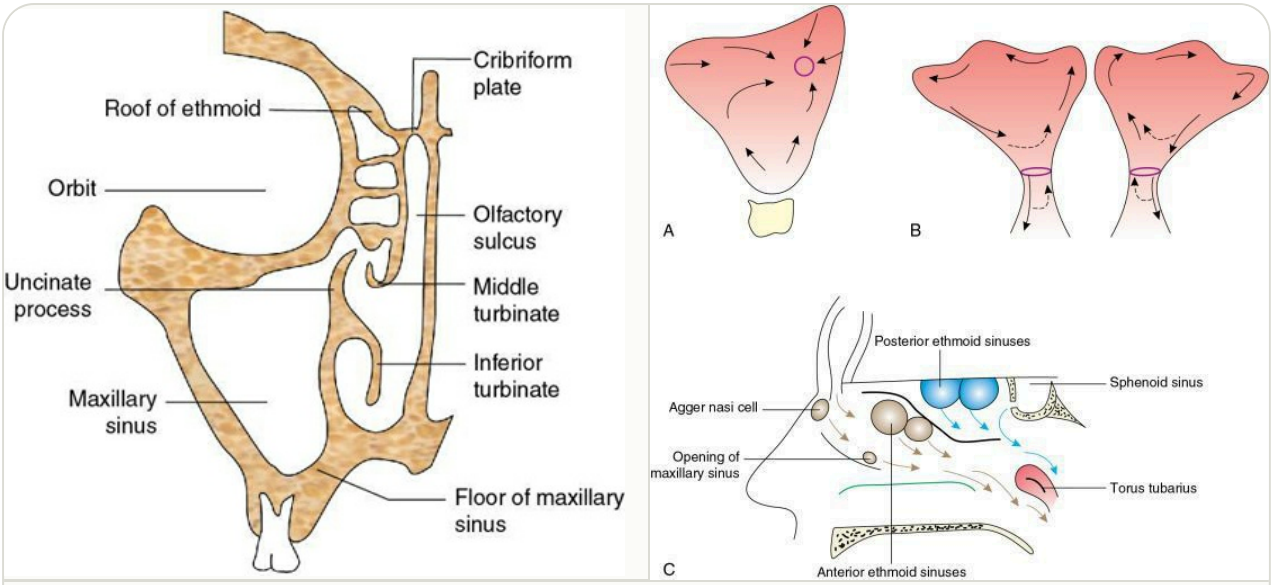


FIGURE 1 First, a coronal section showing the maxillary and ethmoid sinuses in relation to the orbit and nasal cavity. Second, mucociliary clearance patterns of the sinuses.

Dhingra, figs 35.1 and 35.4

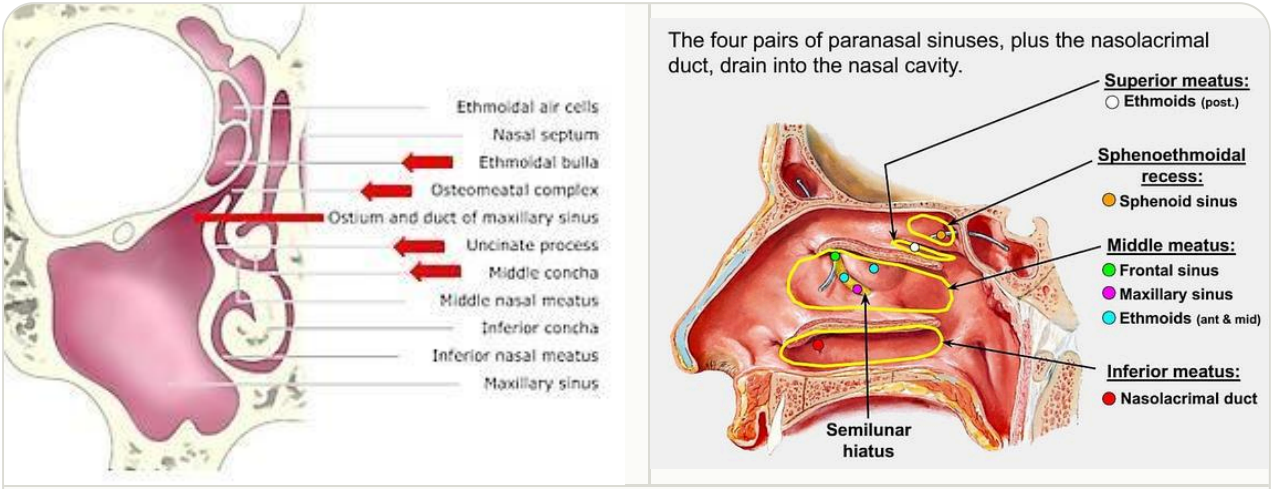


FIGURE 2 The ostiomeatal complex in coronal section, and the drainage of each sinus into its meatus.

Anatomy atlas · ENT Steps p. 108, ENT Steps p. 101

2.2 Development

The maxillary and ethmoid sinuses are present at birth; the frontal sinus appears around the sixth year and the sphenoid later in childhood. So ethmoiditis is the sinusitis of infants, and frontal sinusitis an adult disease.

2.3 The thin walls

The ethmoids are separated from the orbit only by the paper-thin *lamina papyracea*, and the frontal, ethmoid and sphenoid sinuses all border the anterior cranial fossa. This is why complications go to the eye and the brain.

3 CHAPTER THREE Pathogenesis

From virus to bacteria.

3.1 Acute viral rhinosinusitis

Rhinoviruses, influenza and parainfluenza viruses, spread by droplets, with an incubation of one to four days. Congestion, rhinorrhoea, sneezing and low fever improve within seven to ten days.

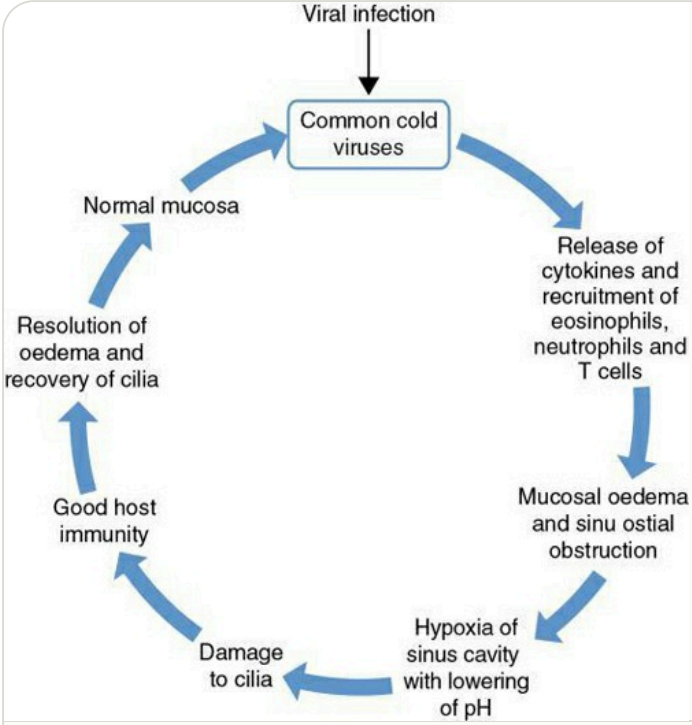


FIGURE 3 Pathophysiology of acute viral rhinosinusitis.
Dhingra, fig. 36.1

3.2 Acute bacterial rhinosinusitis

The virus damages cilia and epithelium; oedema blocks the ostia; secretions stagnate and bacteria multiply: *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Moraxella catarrhalis* and *Staphylococcus aureus*. Anaerobes accompany dental infection.

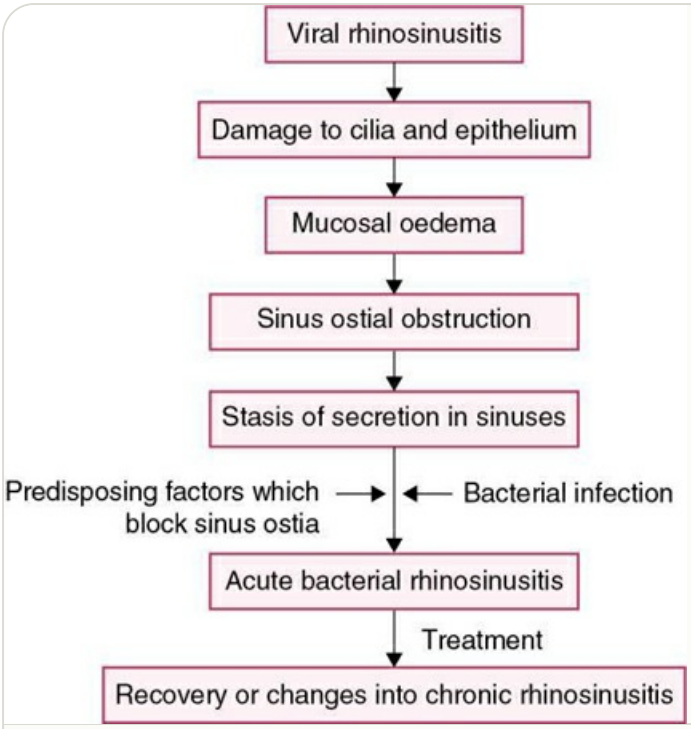


FIGURE 4 The sequence from acute viral to acute bacterial rhinosinusitis.
Dhingra, fig. 36.2

3.3 Diagnosing bacterial infection

Suspect it when symptoms persist beyond ten days without improvement, or worsen after initial improvement (*double worsening*), or with severe symptoms (high fever and purulent discharge) from the start.

3.4 Predisposing factors

Viral infections, swimming and diving, dental infection, trauma, a deviated septum or concha bullosa, polyps, allergy, adenoids, immune deficiency and cystic fibrosis.

4 CHAPTER FOUR Clinical picture by sinus

Where it hurts tells you where it is.

4.1 Acute maxillary sinusitis

4.1.1 Causes

Spread from a cold; swimming in contaminated water; dental infection from the premolar and molar roots, which lie close to the sinus floor, or a root pushed into the sinus, or an oroantral fistula after extraction; trauma.

4.1.2 Clinical features

Fever and malaise; pain over the cheek, often referred to the teeth or gums (so the patient may see a dentist first), worse on stooping, coughing or chewing; tenderness over the anterior wall; redness and swelling of the cheek in children; pus in the middle meatus and postnasal discharge.

4.1.3 Complications

Chronic sinusitis, frontal sinusitis, osteomyelitis of the maxilla, and orbital cellulitis.

4.2 Acute frontal sinusitis

4.2.1 Clinical features

Severe frontal headache with typical periodicity: it starts on waking, peaks around midday and fades by evening (the *office headache*). Exquisite tenderness on pressing upwards on the sinus floor above the medial canthus; upper lid oedema; a streak of pus high in the middle meatus.

4.2.2 Complications

Orbital cellulitis, osteomyelitis of the frontal bone, meningitis, extradural and frontal lobe abscess, and chronic sinusitis.

4.3 Acute ethmoiditis

4.3.1 Clinical features

> The sinusitis of infants and young children. Pain over the bridge of the nose and deep to the eye, worse on eye movement; swelling of both lids; a swollen middle turbinate. Orbital cellulitis comes early.

4.4 Acute sphenoiditis

> ### 4.4.1 Clinical features
>
> Rarely isolated. Headache at the occiput or vertex, sometimes referred to the mastoid, with postnasal discharge seen on the roof of the nasopharynx. Isolated sphenoid opacity needs a mucocele or tumour excluded.

5 CHAPTER FIVE Diagnosis

Clinical first, imaging for complications.

5.1 Examination

Nasal endoscopy shows pus in the middle meatus; a swab from the middle meatus can be cultured.

5.2 Imaging

Not needed for uncomplicated acute disease. CT of the sinuses (with contrast and of the orbits and brain) is requested when a complication is suspected or treatment fails. Plain radiographs (Waters' view) and transillumination are historical.

IMAGING IN THE APP 7 cases

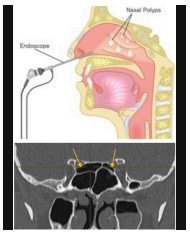
Maxillary, frontal and odontogenic sinusitis, and the orbital and intracranial complications on CT and MRI.



Maxillary Sinusitis
Infection & inflammation · 10 images



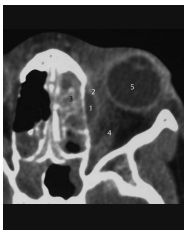
Frontal Sinusitis
Infection & inflammation · 3 images



Sinusitis and its orbital and intracranial complications
Infection & inflammation · 12 images



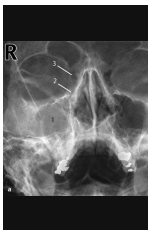
Orbital Cellulitis
Infection & inflammation · 6 images



Subperiosteal Orbital Abscess
Infection & inflammation · 9 images



Intracranial Abscess from a Sinonasal Focus
Infection & inflammation · 2 images



Odontogen Sinusitis
Infection & inflammation 3 images

CHAPTER SIX

6 Management

Most need no antibiotic.

6.1 Viral rhinosinusitis

Symptomatic: saline, analgesics (non-aspirin), short-term decongestants and fluids. Antibiotics are not needed.

6.2 Bacterial rhinosinusitis

- **Analgesics:** NSAIDs or paracetamol.
- **Antibiotics:** amoxicillin, with or without clavulanate, first line; doxycycline or a respiratory quinolone in penicillin allergy. Watchful waiting is reasonable in mild disease with a safety net.
- **Saline irrigation.**
- **Intranasal steroids:** reduce oedema and shorten the illness.
- **Decongestants:** topical for a few days only; oral ones where safe.
- **Antihistamines:** only with coexisting allergy, as they thicken mucus.
- **Steam inhalation:** symptomatic relief.

6.3 Surgery for acute disease

6.3.1 Maxillary

Antral washout is rarely needed now; endoscopic middle meatal antrostomy if drainage is required.

6.3.2 Frontal

If pain, fever or lid swelling worsen despite 48 hours of treatment, *trephine* the frontal sinus through a short incision below the medial end of the eyebrow, take pus for culture, and irrigate through a tube until methylene blue reaches the nose. Endoscopic frontal drainage is the modern alternative.

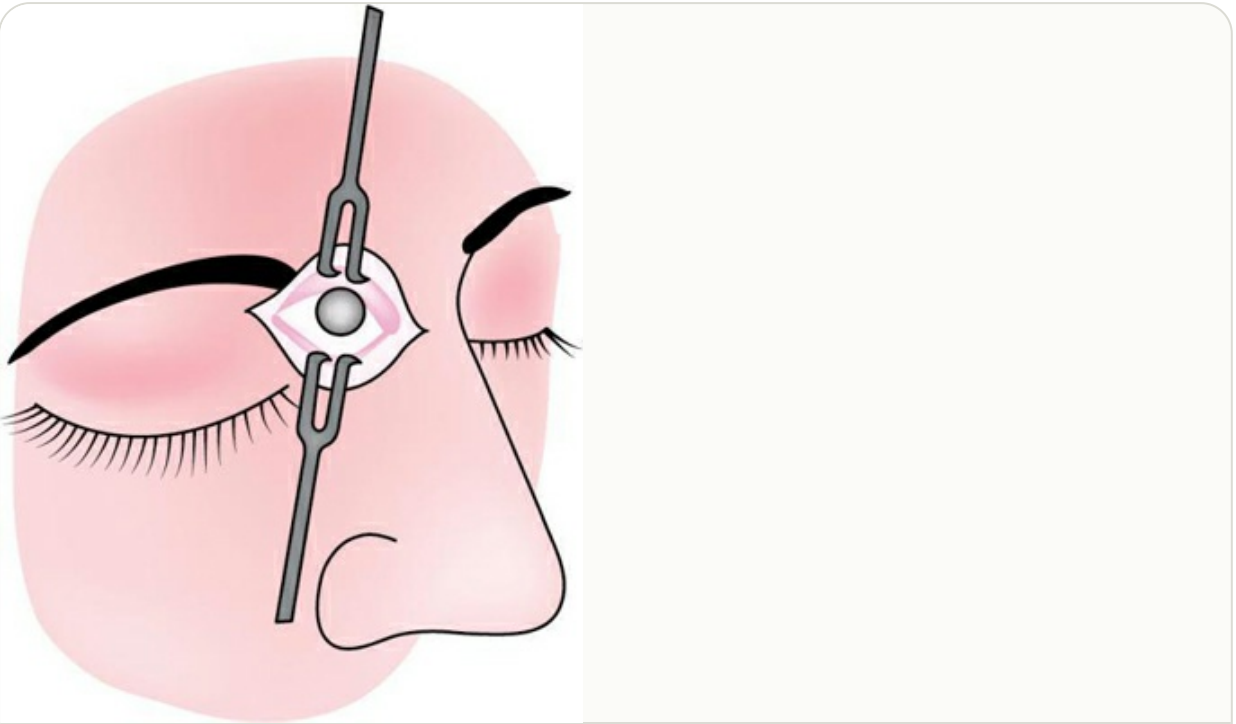


FIGURE 5 Trephination of the right frontal sinus through its floor.
Dhingra, fig. 36.3

6.3.3 Ethmoid

Drain an orbital abscess and the ethmoids, endoscopically or through an external ethmoidectomy.

OPERATIVE GUIDES IN THEATRE 4 operations

Endoscopic Ethmoidectomy
Endoscopic ethmoidectomy

Orbital Abscess
Orbital abscess drainage

Trephination of the Frontal Sinus
Frontal trephination
Frontal Sinus Cranialization
Pott's puffy tumour and cranialisation

7 CHAPTER SEVEN Complications

Local, orbital and intracranial.

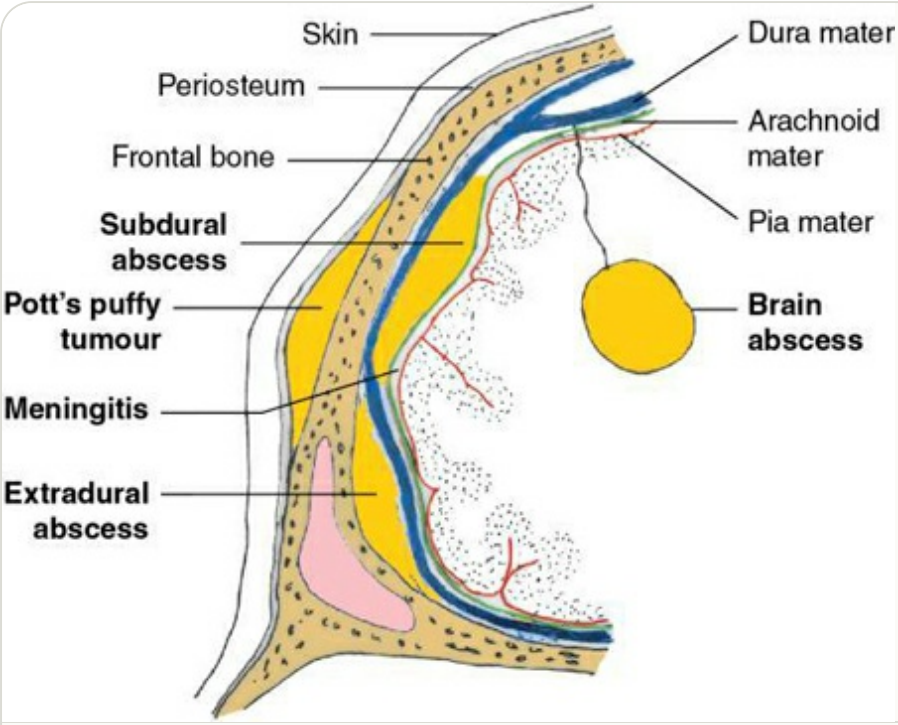


FIGURE 6 Complications of sinusitis.
Dhingra, fig. 38.1

7.1 Local complications

7.1.1 Mucocele and pyocele

An expanding, mucus-filled sinus, from long-standing ostial obstruction, that slowly thins and erodes its bony walls; infected, it is a *pyocele* (mucopyocele).
Frequency: frontal, ethmoid, maxillary, sphenoid.

- **Frontal:** a non-tender cystic swelling above the medial canthus (90%), pushing the eye forward, down and out; sometimes a forehead swelling; egg-shell crackling.
- **Ethmoid:** expands the medial orbital wall and bulges into the middle meatus.

- **Sphenoid:** presents with superior orbital fissure or orbital apex syndrome and proptosis.

Treatment is endoscopic marsupialisation into the nose, which has replaced external operations.

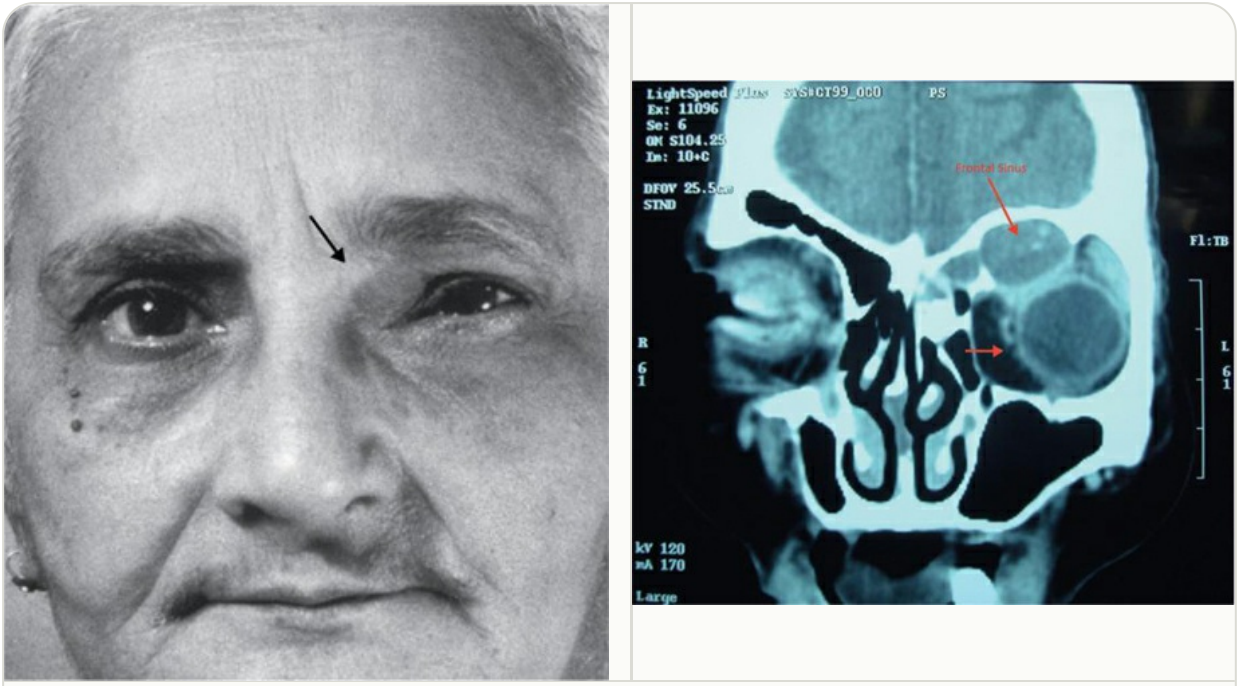


FIGURE 7 First, a frontal mucocele presenting above the left medial canthus. Second, CT of a frontoethmoid mucocele displacing the eye down and out.

Dhingra, figs 38.2 and 38.3

7.1.2 Retention cyst

A cyst of an obstructed mucous gland, usually in the maxillary sinus, found incidentally; it does not erode bone and needs no treatment.

7.1.3 Osteomyelitis

- **Maxilla:** in infants, because the anterior wall contains spongy bone; cheek swelling, lid oedema and fistulae in the infraorbital region, alveolus or palate, with risk to tooth buds.
- **Frontal bone:** in adults; pus under the pericranium forms a soft, doughy forehead swelling, *Pott's puffy tumour*, and often an extradural abscess beneath. Antibiotics, drainage, trephination and removal of sequestra.



FIGURE 8 First, osteomyelitis of the maxilla with an infraorbital fistula. Second, chronic frontal sinusitis with a fistula in the floor of the sinus.

Dhingra, figs 38.4 and 38.5

7.2 Orbital complications

Most come from the ethmoids, through the lamina papyracea or by thrombophlebitis of the ethmoidal veins.

CHANDLER CLASSIFICATION

STAGE	NAME	FEATURES
I	Preseptal cellulitis	Lid oedema; vision and eye movements normal
II	Orbital cellulitis	Diffuse orbital oedema, proptosis, chemosis, restricted movement
III	Subperiosteal abscess	Pus between bone and periosteum; eye displaced away from the abscess
IV	Orbital abscess	Pus within the orbit; severe proptosis, ophthalmoplegia, falling vision
V	Cavernous sinus thrombosis	Spread to the cavernous sinus; signs in both eyes

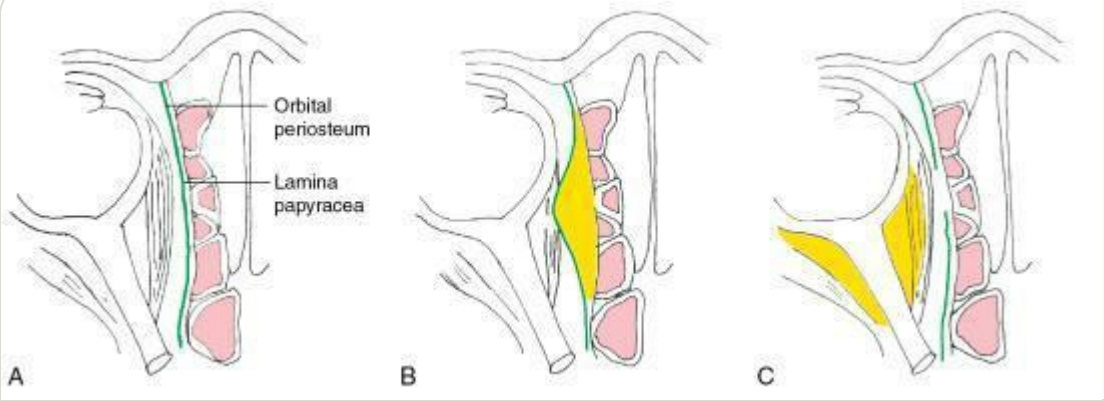


FIGURE 9 Orbital complications: (A) normal, (B) subperiosteal abscess, (C) orbital abscess. Dhingra, fig. 38.7

7.2.1 Displacement by the source

An ethmoid subperiosteal abscess pushes the eye forward, down and out; a frontal one, down and out; a maxillary one, up and forward.

7.2.2 Orbital apex and superior orbital fissure syndromes

Posterior ethmoid or sphenoid infection: deep orbital pain and paralysis of the sixth, third and fourth nerves (superior orbital fissure syndrome), with loss of vision and of V2 sensation in the orbital apex syndrome.

THE SWOLLEN EYE

- Check visual acuity, colour vision, pupils, eye movements and proptosis.
- Any loss of vision, colour vision or movement, or a relative afferent pupil defect: urgent contrast CT and surgical drainage.
- Involve ophthalmology early; examine both eyes for cavernous sinus spread.

7.2.3 Treatment

Admission, intravenous antibiotics and nasal decongestion; contrast CT. Drain a subperiosteal or orbital abscess (endoscopically or externally) when vision is threatened, there is no improvement in 24 to 48 hours, or the abscess is large.

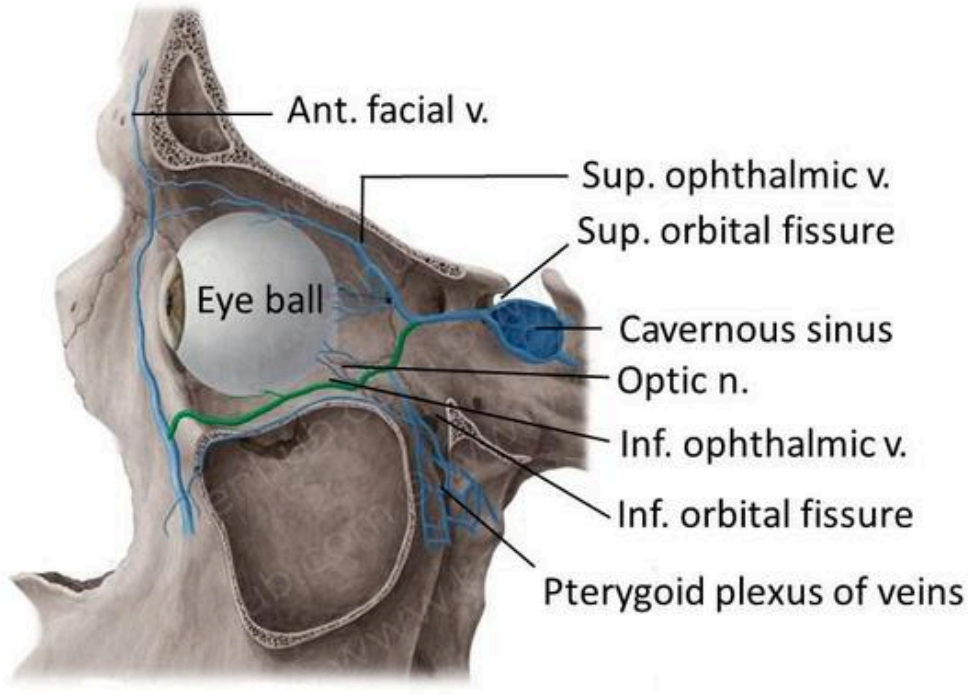


FIGURE 10 Orbital veins draining to the cavernous sinus and pterygoid plexus: the valveless route for spread.

Anatomy atlas · Doss p. 101

7.3 Intracranial complications

Meningitis, extradural, subdural and brain (frontal lobe) abscess, and cavernous sinus thrombosis, from the frontal, ethmoid and sphenoid sinuses.

7.3.1 Cavernous sinus thrombosis

Spread through the valveless ophthalmic veins from the ethmoids, sphenoid, orbit, the danger area of the face, or the pharynx. Abrupt onset with rigors; both eyes become involved; chemosis and proptosis; sequential palsies of the third, fourth and sixth nerves up to total ophthalmoplegia; a fixed, dilated pupil; reduced V1 sensation; papilloedema. Diagnosed with contrast CT or MRI.

ORBITAL CELLULITIS OR CAVERNOUS SINUS THROMBOSIS?

	ORBITAL CELLULITIS	CAVERNOUS SINUS THROMBOSIS
Source	Usually ethmoid sinus	Nose, sinuses, orbit, face, pharynx, ear
Onset	Gradual	Abrupt, with rigors
Eyes	One	Spreads to both

	ORBITAL CELLULITIS	CAVERNOUS SINUS THROMBOSIS
Chemosis and proptosis	Moderate	Marked
Cranial nerves III, IV, VI	Movement limited by swelling	Progressive palsies
Pupil	Usually normal	Dilated, fixed
General state	Unwell	Toxic, very ill

7.4 Descending and focal infections

Infected discharge running down the pharynx causes pharyngitis, laryngitis and bronchitis; sinus infection can also act as a focus for distant disease.

EPOS 2020 AND ANTIBIOTIC STEWARDSHIP (OXFORD TEXTBOOK)

EPOS 2020 defines three phases of acute rhinosinusitis (symptoms for less than 12 weeks). The **common cold** (acute viral rhinosinusitis) lasts less than 10 days. **Acute post-viral rhinosinusitis** is when symptoms worsen after five days or persist beyond ten. **Acute bacterial rhinosinusitis** is suggested by at least three of: discoloured discharge (often one-sided), severe local pain (often one-sided), fever above 38 °C, raised ESR or CRP, and "double sickening" (worsening after an initial improvement). Only 0.5 to 2% of viral episodes become bacterial, and discoloured mucus alone does not mean bacteria. EPOS 2020 dropped the term "subacute". **Recurrent acute rhinosinusitis** means four or more episodes a year with symptom-free intervals.

Antibiotics are greatly overused, given on the first day in many episodes and to over 90% of patients in UK primary care, although early antibiotics do not prevent complications. Most patients need only analgesia, saline and, for post-viral symptoms, a nasal steroid spray; antibiotics are kept for severe or bacterial disease. The usual organisms remain *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Moraxella catarrhalis*, but resistance to penicillin and macrolides has risen sharply. *Streptococcus anginosus* (milleri group) is the commonest organism in complicated sinusitis in children and adolescents, with a tendency to form intracranial abscesses.

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 On call

SWOLLEN EYE WITH SINUSITIS

1. Check vision, colour vision, pupils, movements and proptosis, and examine the other eye.
2. Admit; start intravenous antibiotics and nasal decongestion.
3. Contrast CT of the sinuses, orbits and brain.
4. Involve ophthalmology; drain an abscess if vision is threatened or there is no improvement.
5. Headache, drowsiness or meningism: image the brain and involve neurosurgery.

8.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definitions** by duration, and when to diagnose bacterial infection.
- **Organisms** and first-line antibiotic.
- **Features of each sinus,** including the office headache.
- **Frontal trephination.**
- **Chandler's classification** and the direction of eye displacement.
- **Mucocele and Pott's puffy tumour.**
- **Orbital cellulitis versus cavernous sinus thrombosis.**

8.3 Common pitfalls

COMMON PITFALLS

- Antibiotics for every cold.
- Calling an orbital abscess preseptal cellulitis without checking vision and movement.
- Missing the extradural abscess beneath a Pott's puffy tumour.

Written for this app from Oxford Textbook of Otolaryngology ch. 36, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 9, Dhingra ch. 36, 38 and K. J. Lee ch. 28. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Maxillary Sinusitis

IMAGING FINDINGS

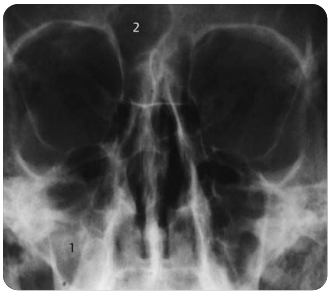
- Air–fluid level, frothy or bubbly secretions
- Mucosal thickening
- Chronic: sclerotic thickened walls (osteitis)

TOP DIFFERENTIALS

- Retention cyst or polyp
- Post-traumatic blood
- Odontogenic source (periapical lucency, oroantral fistula)

CHECKLIST

- Look at the teeth and the ostiomeatal unit
- Contrast CT only for complications



Caldwell view. This 26-year-old patient had pain in the right maxillary region. The right maxillary sinus seems to be opacified in the lower part (1). The lef…



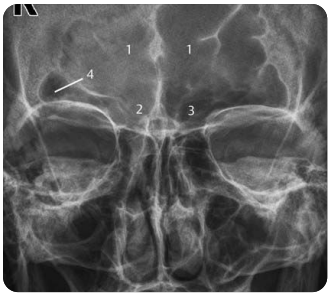
Axial NECT in patient with purulent nasal discharge and headache shows patchy bilateral ethmoid sinus disease, an air-fluid level in the right sphenoid…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 688.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Frontal Sinusitis

- The anatomy and pathology of the frontal recess required special attention.
- In cases of persisting drainage problems in the region of the frontal recess, the septum between the frontal sinuses may be removed for establishing drainage by way of the left frontal recess.
- Bony destruction with infiltrative characteristics is suggestive of malignancy.
- Calcifications in the opacified area are suggestive of fungal infections.



Plain radiograph, Caldwell. It is difficult to evaluate the superior aspect of the frontal sinuses (1) to determine whether these are opacified or are…

PARANASAL SINUSES · INFECTION & INFLAMMATION

Sinusitis and its orbital and intracranial complications

IMAGING FINDINGS

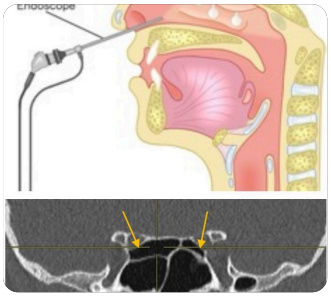
- Orbit: preseptal or postseptal stranding, subperiosteal or orbital abscess
- Intracranial: epidural or subdural empyema, abscess, meningitis
- Venous: superior ophthalmic vein and cavernous sinus thrombosis

TOP DIFFERENTIALS

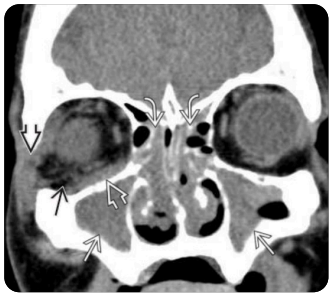
- Idiopathic orbital inflammation
- Tolosa-Hunt syndrome

CHECKLIST

- Contrast CT for the orbit, MRI for intracranial disease
- Frontal sinusitis: look for Pott puffy tumour



Coronal reconstructed image from a high-resolution CT. Bilateral Onodi cells (arrows). This puts the optic nerves at risk of damage during FESS…



Coronal NECT in a man with acute sinusitis complicated by orbital cellulitis and subperiosteal abscess shows extensive opacification of…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 696.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Orbital Cellulitis

IMAGING FINDINGS

- Stranding of the postseptal fat
- Swollen extraocular muscles; proptosis
- No discrete collection

TOP DIFFERENTIALS

- Subperiosteal abscess
- Invasive fungal infection
- Idiopathic orbital inflammation

CHECKLIST

- Separate preseptal from postseptal
- Repeat imaging if not improving



CT, axial with intravenous contrast. Bilateral opacification of the ethmoid (1). The lamina papyracea is intact without signs of periosteal…



Axial CECT in a patient with facial impetigo shows edema and enhancement of the preseptal soft tissues with normal appearance of th…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 814.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Subperiosteal Orbital Abscess

IMAGING FINDINGS

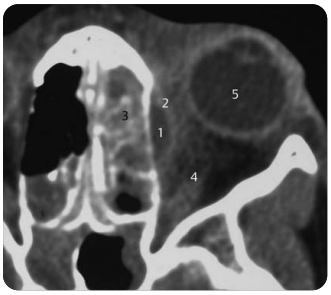
- Lentiform rim-enhancing collection along the lamina papyracea
- Medial rectus and globe displaced laterally
- Adjacent ethmoid opacification; diffusion restriction

TOP DIFFERENTIALS

- Orbital cellulitis (phlegmon)
- Idiopathic orbital inflammation
- Mucocele

CHECKLIST

- Size, posterior extent and vision
- Superior ophthalmic vein and cavernous sinus



CT, axial with intravenous contrast enhancement. The CT scan shows a subperiosteal orbital abscess (1) with typical rim enhancement (2). The…



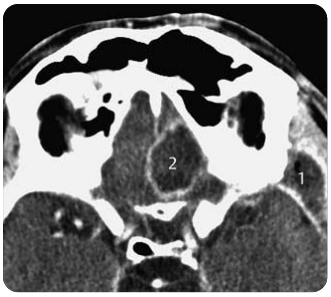
Axial CECT shows asymmetric opacification of the left ethmoid sinuses with a large subperiosteal abscess extending into the medial…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 810.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Intracranial Abscess from a Sinonasal Focus

- Frontal and ethmoid sinusitis spread through valveless diploic veins or bone erosion to cause epidural, subdural or frontal lobe abscess (Pott's puffy tumour is the extracranial sign).
- Contrast MRI is more sensitive than CT for subdural empyema and cerebritis; repeat imaging if seizures or reduced consciousness develop after sinus surgery.
- Combined management: neurosurgical drainage with endoscopic sinus surgery to drain the source.

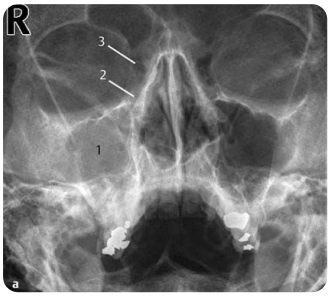


CT with intravenous contrast, axial. After initial antibiotic therapy, a subdural empyema (1) was drained surgically. In the basal left frontal lobe…

PARANASAL SINUSES · INFECTION & INFLAMMATION

Odontogenic Sinusitis

- Destruction of bony outlines with infiltrative characteristics is suggestive of inverted papilloma, malignancies, or fungal infections.
- Calcifications in the opacified area are suggestive of fungal infections.
- History of frequent dental consultations and resorption of bone around tooth roots may indicate odontogenic infections.
- Long-term antibiotics are an essential part of postoperative treatment.



Plain radiograph, Waters view. There is opacification of the right maxillary sinus (1) as well as in the region of the right ethmoid (2). The right frontal…